

MUHAMMAD ANEES

DATA SCIENCE | ARTIFICIAL INTELLIGENCE | MACHINE LEARNING | DEEP LEARNING | XR/VR/AR/MR

Islamabad, Pakistan | +92 335 742 8855 | animalik272@gmail.com | LinkedIn | Portfolio

OBJECTIVE

To pursue a PhD in Data Science, Artificial Intelligence, Extended Reality (XR), or a closely related interdisciplinary field where I can develop rigorous, impactful research at the intersection of data-driven intelligence, immersive technologies and real-world applications. I am particularly motivated by machine learning, deep learning, NLP/LLMs, generative AI, predictive analytics, computer vision and intelligent systems, with healthcare AI as one important application area and a growing focus on integrating AI with Virtual Reality (VR), Augmented Reality (AR) and Mixed Reality (MR). My goal is to contribute to a strong research group, publish high-quality scientific work, and develop reliable, explainable and practically useful AI/XR solutions.

RESEARCH PROFILE

Data Science and Artificial Intelligence professional with an MS in Data Science and a BS in Information Technology. Research experience spans machine learning, deep learning, natural language processing, predictive analytics and intelligent systems, including applied work in healthcare and drug-drug interaction prediction. Complemented by 5+ years of professional software and immersive-technology development, including Unity-based VR, AR, MR, WebXR, Meta Quest and interactive 3D systems. Strong interest in intelligent and adaptive XR, multimodal interaction, predictive user modelling, Human-Computer Interaction, and data-driven approaches to real-world problems.

RESEARCH INTERESTS

Data Science • Artificial Intelligence • Machine Learning • Deep Learning • Natural Language Processing • Large Language Models (LLMs) • Generative AI • Predictive Analytics • Computer Vision • Intelligent Systems • Healthcare AI • Extended Reality (XR) • Virtual Reality (VR) • Augmented Reality (AR) • Mixed Reality (MR) • Multimodal Interaction • Human-Computer Interaction • Intelligent & Adaptive XR • AI + XR

EDUCATION & THESIS

MS Data Science

Bahria University, Islamabad, Pakistan | 2024 | CGPA 3.20/4.00

Master's Thesis: *Amend Complementary and Integrative Health Approaches for Untangling Prevalent Health Issues*

- Applied data-science and machine-learning methods to a real-world healthcare prediction and recommendation problem.
- Evaluated SVC, Random Forest, Gradient Boosting, KNN, Multinomial Naïve Bayes and Neural Network models.
- Dataset scope: 41 diseases and 133 symptoms; reported top accuracy of 97.62%.

Repository: Bahria University DSpace · Thesis Record

BS Information Technology

International Islamic University Islamabad, Pakistan | 2017-2021 | CGPA 3.52/4.00

Final Year Project: *Smart Builders House Estimation & Care App*

- React Native and Firebase mobile application.
- Third Position in BSIT; Distinct Student Award.

RESEARCH PUBLICATIONS — SUBMITTED

Amend Complementary and Integrative Health Approaches for Untangling Prevalent Health Issues

Muhammad Anees, Madeeha Asif, & Muhammad Asfand E Yar | Submitted manuscript

- Data-science study integrating symptom-diagnosis data with a self-created CIH dataset for prediction and recommendation.
- Models: SVC, Random Forest, Gradient Boosting, KNN, Multinomial Naïve Bayes and Neural Network; reported best accuracy 97.62%.

Suggestive Strategies for Mitigating Adverse Effects in Drug-Drug Interactions

Madeeha Asif, Muhammad Anees, & Muhammad Asfand E Yar | Submitted manuscript

- Deep-learning prediction of 65 DDI-related event types using DrugBank data.
- Features include SMILES molecular structures, drug targets, enzymes and pathways, with NLP-based processing and Jaccard similarity.
- Reported CNN performance: 89.89% accuracy, 87.11% precision, 77.23% recall and 80.72% F1-score.

RESEARCH & TECHNICAL PROJECTS

Healthcare ML / Decision-Support Research

MS Data Science Thesis | 2024

- Symptom-based disease prediction and recommendation using comparative machine-learning evaluation.

Drug-Drug Interaction Prediction

Research Manuscript | Submitted

- Deep-learning DDI prediction using DrugBank-derived molecular, target, enzyme and pathway features.

Smart Builders House Estimation & Care App

BSIT Final Year Project | 2021

- React Native + Firebase application for construction-related estimation and management.

Immersive Computing / Intelligent XR Projects

Professional / Independent | 2021-Present

- VR, AR, MR, WebXR, multiplayer and interactive applications using Unity, Meta Quest and related technologies.
- XR interaction systems, immersive 3D environments, APIs, prototyping, performance optimization and deployment.
- Research interest in combining AI/ML, multimodal user data and adaptive interaction with immersive XR systems.

PROFESSIONAL EXPERIENCE

VR / AR Developer

Quantum Edge LLC, Pakistan | June 2025 - present

- Lead development of VR, AR and immersive applications in Unity; scripting, interaction systems, animation, APIs, prototyping and optimization.
- Design, test and publish applications for Meta Quest and mobile platforms.

VR / MR Developer

FI Consultancy, Islamabad, Pakistan | May 2023 - June 2025

- Developed VR and Mixed Reality applications using Unity, including immersive environments, interactions, animation, scripting and optimization.
- Debugging, performance optimization, feature implementation and Meta Quest deployment.

Co-Lead Game Developer

Koncept Solutions International, Rawalpindi, Pakistan | April 2022 - June 2023

- Co-led Unity game development covering gameplay systems, environments, animation, scripting, prototyping and optimization.
- Technical problem solving and project implementation.

Teacher

APS Chagmalai SWTD, Pakistan | August 2021 - February 2022

- Taught Computer Science, Mathematics and Science; supported programming and analytical learning.

React Native Intern

Enabling Systems (Pvt.) Ltd., Rawalpindi, Pakistan | January 2021 - August 2021

- Developed a construction-related mobile application using React Native and Firebase.

WordPress Developer / Freelancer

Independent | 2017 - 2020

• Developed client websites using WordPress and gained practical web-development experience.

TECHNICAL SKILLS

AI / ML: Machine Learning • Deep Learning • CNN • DNN • SVC/SVM • Random Forest • Gradient Boosting • KNN • Naïve Bayes • Neural Networks

Data Science: Data preprocessing • Model training • Feature selection • Predictive analytics • Data analysis • Data visualization • Power BI

NLP / Research: Natural Language Processing • Healthcare AI • Pattern recognition • Intelligent systems • Research methodology • Model evaluation

XR / Immersive Computing: Unity • VR • AR • MR • XR • WebXR • Meta Quest • HoloLens • Immersive interaction • Interactive 3D environments • XR prototyping • Performance optimization • Deployment

Programming / Development: Python • C# • JavaScript • React Native • Firebase • APIs • Version control

AI + XR / HCI: Intelligent & adaptive XR • Multimodal interaction • Predictive user modelling • Human-Computer Interaction • AI-enabled immersive systems

AWARDS & ACADEMIC ACHIEVEMENTS

• Employee of the Month — FI Consultancy, 2024.
• Employee of the Month — Koncept Solutions International, 2023.
• Best Teacher Award — APS Chagmalai SWTD, 2022.
• Third Position Holder in BSIT — International Islamic University Islamabad, 2021.
• Distinct Student Award — International Islamic University Islamabad, 2021.

TRAINING, LEADERSHIP & SERVICE

• International Training Workshop on Programming, Mobile App and Website Development — IIUI, October 2020.
• Seminar on Artificial Intelligence and Machine Learning — Bahria University Islamabad, 2023.
• Member & Polio Plus Chair, Rotaract Club of Rawalpindi City — 2021-2023.
• Volunteer — Qalam Movement; Taleem o Tarbiat Foundation; Musa Khar Welfare Trust.

DIGITAL PROFILES

Portfolio: https://aneesvr016.github.io/My-Profile-/

LinkedIn: https://www.linkedin.com/in/muhammad-anees-b72404210

Thesis: Bahria University DSpace Thesis Record

LANGUAGES & REFERENCES

English — Professional proficiency | Urdu — Professional proficiency | Saraiki — Professional proficiency.